

NMAM INSTITUTE OF TECHNOLOGY, NITTE*(An Autonomous Institution affiliated to VTU, Belagavi)***Fifth Semester B.E. (Biotechnology) (Credit System) Degree Examinations**

January 2023

BT*112 – RESEARCH METHODOLOGY

Duration: 3 Hours

Max. Marks: 100

Note: 1) Answer **Five full** questions choosing **Two full** questions from **Unit – I & Unit – II** each and **One full** question from **Unit – III**.

2) Use of **Statistical Tables** is allowed

Unit – I		Marks	BT*	CO*	PO*
1.	a) What do you mean by research methods? Give two examples. What are the postulates on which the scientific methods are based?	10	L*2	1	1
	b) How are questionnaires suitably used for primary data collection? Give its merits and demerits.	10	L1	2	4
2.	a) How would you classify research on the basis of objectives? Explain with one example each.	10	L2	1	1
	b) Why is editing of data essential for survey-based approaches? Explain the types of editing with respect to questionnaires and interviews.	10	L2	2	4
3.	a) Give an account on the principles of experimental design with examples.	10	L1	1	3
	b) What are the measurement scales that are applied for qualitative data? Explain in detail with examples.	10	L2	2	2
Unit – II					
4.	a) Hypothesis is a mere assumption or some supposition to be proved or disproved. Explain with examples. What are the characteristics of hypothesis?	10	L2	3	2
	b) What are the sequential steps in conducting and analyzing experiments on the basis of DoE? Explain in detail.	10	L2	4	4
5.	a) The specimen of copper wires drawn from a large lot have the following breaking strength (in kg weight). 578,572,570,568,572,578,570,572,596,544 Test using Students t-statistic, whether the mean breaking strength of the lot may be taken to be 578 kg. weight (Test at 5% level of significance).	10	L3	3	2
	b) Which experimental design could be suitably used for screening of more than 5 factors? Elaborate on the design.	10	L2	4	4
6.	a) What is ANOVA? Explain its basic principle.	10	L1	3	1
	b) How would you analyse the results obtained from optimization by response surface methodology? Explain.	10	L2	4	4
Unit – III					
7.	a) Explain in detail the steps of writing a report. Also give an account on the necessary precautions to be followed.	10	L1	5	10
	b) What is plagiarism and self-plagiarism? Explain their types	10	L2	5	8
8.	a) What is the meaning of interpretation? What are the precautions to be taken during interpretation of results?	10	L2	5	10
	b) Differentiate between technical and popular report. Also explain the layout of a technical report	10	L2	5	10

BT* Bloom's Taxonomy, L* Level; CO* Course Outcome; PO* Program Outcome
